

→ /*eq1:[2*x+3*y=8,x-y=1];*/;

(%i10) **A:matrix([2,3,8],[1,-1,1]);**

A
$$\begin{bmatrix} 2 & 3 & 8 \\ 1 & -1 & 1 \end{bmatrix}$$

(%i11) **echelon(A);**

(%o11)
$$\begin{bmatrix} 1 & \frac{3}{2} & 4 \\ 0 & 1 & \frac{6}{5} \end{bmatrix}$$

(%i12) **linsolve([x+(3/2)*y=4,y=(6/5)],[x,y]);**

(%o12)
$$\left[x = \frac{11}{5}, y = \frac{6}{5} \right]$$

→ /*eq2:[3*x-2*y=7,x+y=4]*/;

(%i13) **B:matrix([3,-2,7],[1,1,4]);**

B
$$\begin{bmatrix} 3 & -2 & 7 \\ 1 & 1 & 4 \end{bmatrix}$$

(%i14) **echelon(B);**

(%o14)
$$\begin{bmatrix} 1 & -\left(\frac{2}{3}\right) & \frac{7}{3} \\ 0 & 1 & 1 \end{bmatrix}$$

(%i15) **linsolve([x-(2/3)*y=(7/3),y=1],[x,y]);**

(%o15)
$$[x = 3, y = 1]$$

→ /*eq3:[x+y+z=6,2*x-y+z=3,x+2*y-z=3]*/;

(%i20) **C:matrix([1,1,1,6],[2,-1,1,3],[1,2,-1,3]);**

C
$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 3 \end{bmatrix}$$

(%i22) **echelon(C);**

(%o22)
$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & \frac{1}{3} & 3 \\ 0 & 0 & 1 & \frac{18}{7} \end{bmatrix}$$

(%i23) **linsolve([x+y+z=6,y+(1/3)*z=3,z=18/7],[x,y,z]);**

(%o23)
$$\left[x = \frac{9}{7}, y = \frac{15}{7}, z = \frac{18}{7} \right]$$

→ /*eq4:[2*x+y-z=8,-x+3*y+2*z=3,3*x+2*y+z=13]*/;

(%i25) **D:matrix([2,1,-1,8],[-1,3,2,3],[3,2,1,13]);**

D
$$\begin{bmatrix} 2 & 1 & -1 & 8 \\ -1 & 3 & 2 & 3 \\ 3 & 2 & 1 & 13 \end{bmatrix}$$

(%i26) **echelon(D);**

(%o26)
$$\begin{bmatrix} 1 & \frac{1}{2} & -\left(\frac{1}{2}\right) & 4 \\ 0 & 1 & \frac{3}{7} & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

(%i28) **linsolve([x+(1/2)·y-(1/2)·z=4,y+(3/7)·z=2,z=0],[x,y,z]);**

(%o28) [x=3,y=2,z=0]

→ /*eq5:[x+2·y+3·z=14,2·x-y+z=5,3·x+y+2·z=10]*/;

(%i29) **E:matrix([1,2,3,14],[2,-1,1,5],[3,1,2,10]);**

E
$$\begin{bmatrix} 1 & 2 & 3 & 14 \\ 2 & -1 & 1 & 5 \\ 3 & 1 & 2 & 10 \end{bmatrix}$$

(%i30) **echelon(E);**

(%o30)
$$\begin{bmatrix} 1 & 2 & 3 & 14 \\ 0 & 1 & 1 & \frac{23}{5} \\ 0 & 0 & 1 & \frac{9}{2} \end{bmatrix}$$

(%i32) **linsolve([x+2·y+3·z=14,y+z=23/5,z=9/2],[x,y,z]);**

(%o32)
$$\left[x = \frac{3}{10}, y = \frac{1}{10}, z = \frac{9}{2} \right]$$